Supervised Learning

01

What are Supervised Learning Algorithms?

Supervised learning algorithms are machine learning methods where the model is trained using labeled data (input + correct output). The model learns the relationship between input and output to make predictions on new data. Examples include regression and classification algorithms.

02

Difference between Regression vs Classification.

Regression predicts **continuous values** (e.g., house price, salary), while classification predicts **categorical labels** (e.g., yes/no, spam/not spam). Regression output is numeric, whereas classification output is discrete classes.

03

Explain Simple Linear Regression

Simple Linear Regression is a technique used to model the relationship between one independent variable (X) and one dependent variable (y). It fits a straight line equation:

$$y=mx+c$$

where m is slope and c is intercept.

04

Assumptions of Linear Regression

- Linear relationship between variables
- Independence of errors
- Constant variance (Homoscedasticity)
- No multicollinearity
- Errors are normally distributed

These assumptions ensure accurate and reliable predictions.

05

What is Bias–Variance Trade-Off?

Bias–Variance Trade-Off is the balance between underfitting and overfitting.

- High bias → model too simple (underfitting)
- High variance → model too complex (overfitting)
  Goal is to find a model that balances both for best performance.

06

Overfitting vs Underfitting

- **Underfitting:** Model is too simple and cannot capture patterns (e.g., straight line on curved data)
- **Overfitting:** Model learns noise and performs well on training but poorly on test data

Example: Polynomial regression with very high degree → overfitting, while linear model on curved data → underfitting.